MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 12

Power BI - Connecting To Data Using Power Query

1. INTRODUCTION

MS Excel can be used to convert raw data into meaningful visualizations. Although excel has its own limitations, there are add-ins that make excel an excellent path for data visualization.

1.1 Learning Objective:

- *Understand the various add-ins in the Excel BI toolkit to import, model, prepare, and visualize data.*

Connecting to Excel Files.

We need to learn how to source, load, and modify the data before we enter into a visualization of the data. Here we are required to understand:

- Data discovery
- Data loading
- Data modification

Data Discovery

Data discovery is the process of finding data sources by connecting to various data sources which include:

- Relational data
- Structured data
- Semi-structured data
- Nonstructured data

Data Loading

Data loading is the process of loading data from data sources into excel for analysis.

Data Modification

Data modification helps in understanding how to,

- Modify data and filter it.
- Join separate data structures.

The combined application of data discovery, data loading and data modification is also called as ETL (Extract, Transform , Load) data.

For earlier versions of Excel (before 2016) power query was an add-in available to discover, access, and consolidate information from various sources. Power Query now in updated version is known as Get & Transform features which helps with the process of collecting, combining, and refining data sources. The four phases of Power Query are: -

- Connect
- Transform

- Combine
- Load

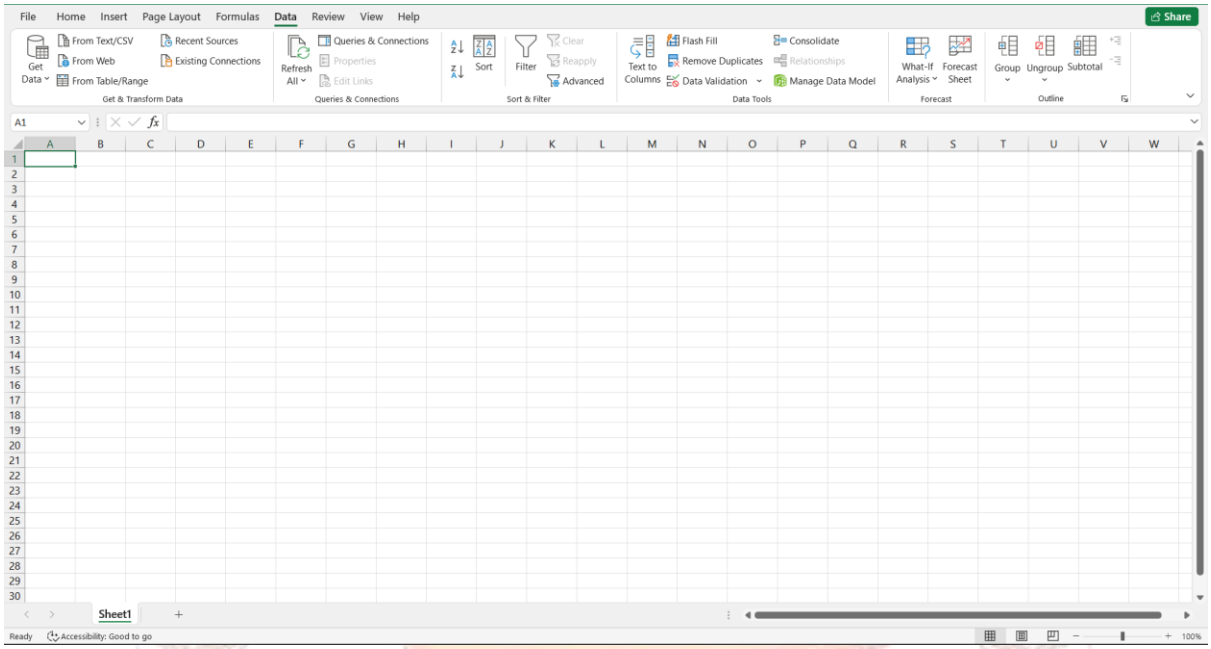


Fig 1: Excel File

Figure 1 shows a blank excel worksheet. Power Query (also known as Get & Transform in Excel) can help in importing or connecting to external data and then shaping that data according to requirement, and then loading the query into Excel to create charts and reports.

Let's look at the process of connecting the data. Using the get and transform utility we can connect to file data such as Excel, CSV, XML, text, JSON, etc as shown in figure 2.

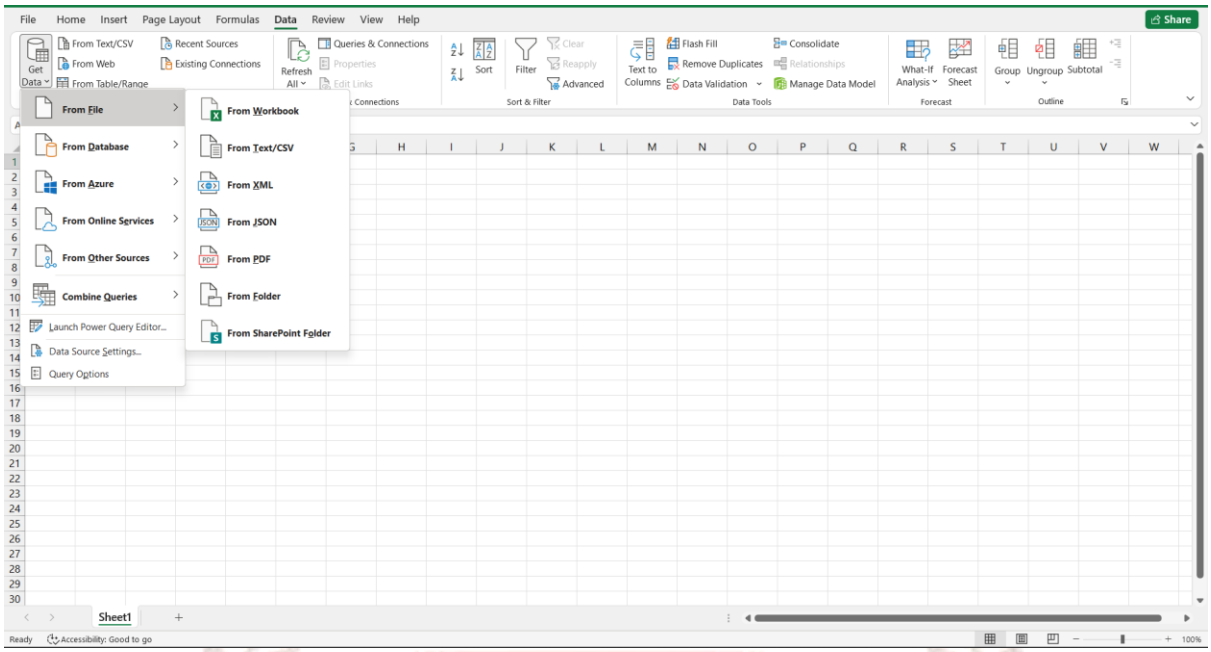


Fig 2: File Data Connection

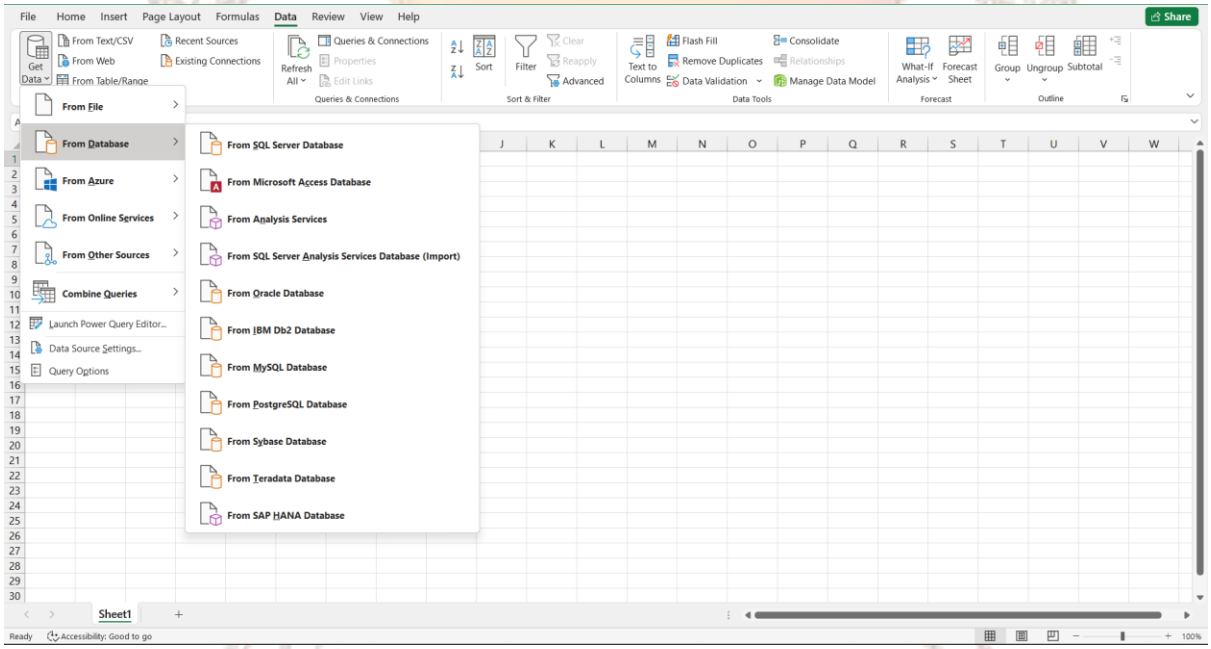


Fig 3: Database Connection

Figure 3 shows we can also connect to databases such as SQL server database, Microsoft Access database, Oracle database, Sybase database, SAP Hana database, and so on. Similarly, the data connection is possible with different sources of Azure storage and their databases.

Data collection from online services like SharePoint online list, Microsoft Exchange, Facebook, and salesforce can be used to connect as a new query.

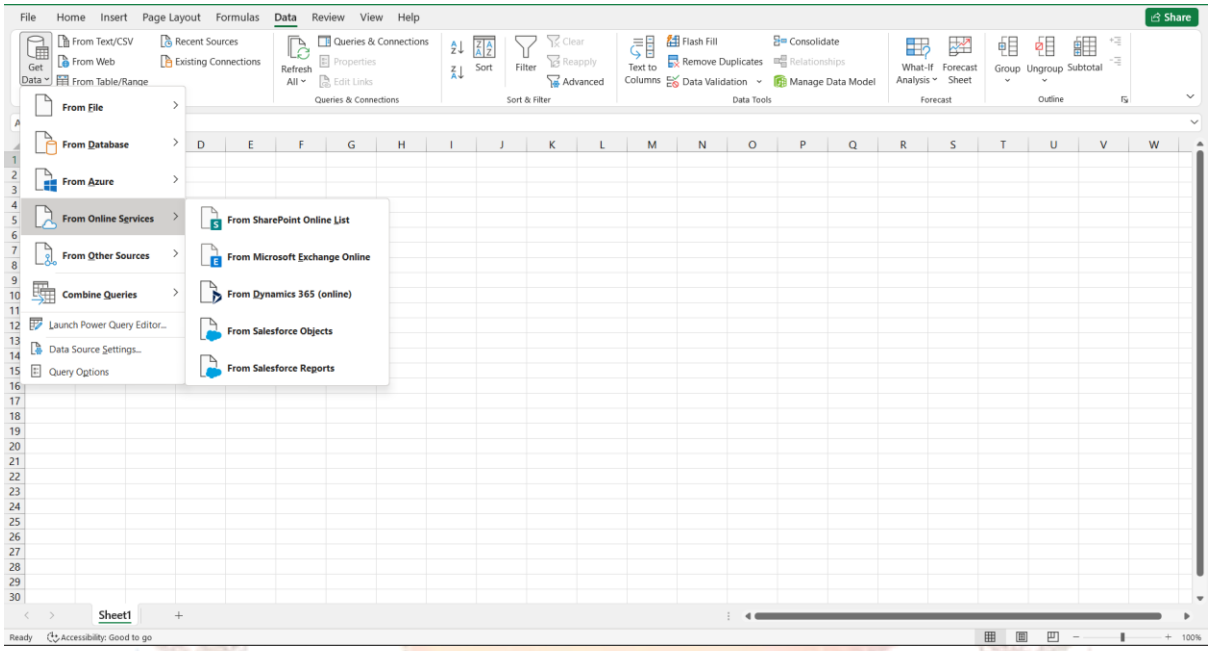


Fig 4: Data from Online Services

Data from other sources such as web page, SharePoint list, OData feed, Hadoop file, ODBC service are used to connect external data to active directory.

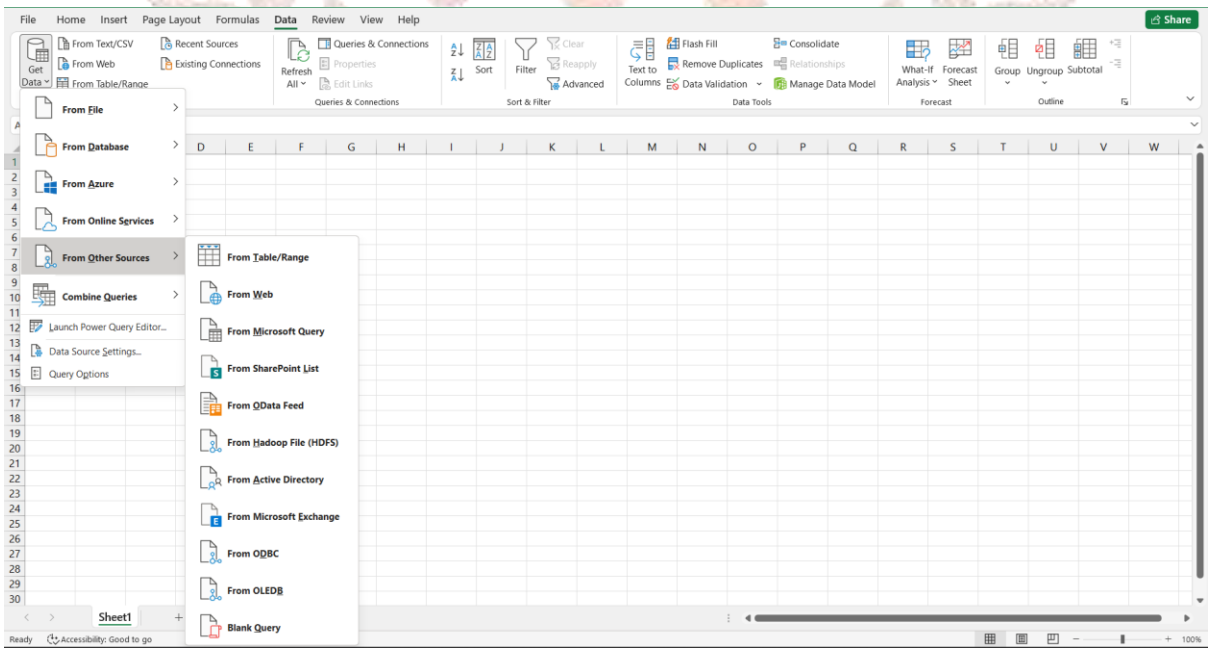


Fig 5: Data from Other Services

Connecting to File Data

Let's look at the process of connecting to a file or an excel workbook.

First, select the new query option in the data tab and then click on from file option, then click on from Excel workbook, and then select which workbook needs to open from the database as shown in figure 6.

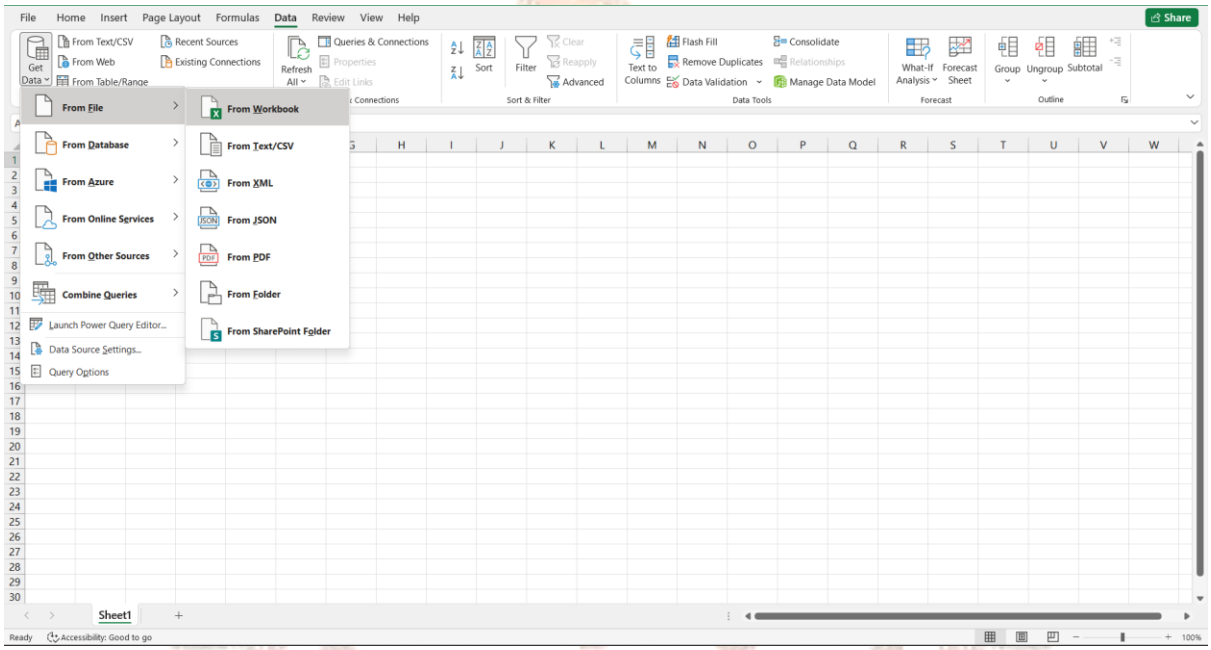


Fig 6: Connecting to Data from Workbook

Here we are considering the hospital patient data

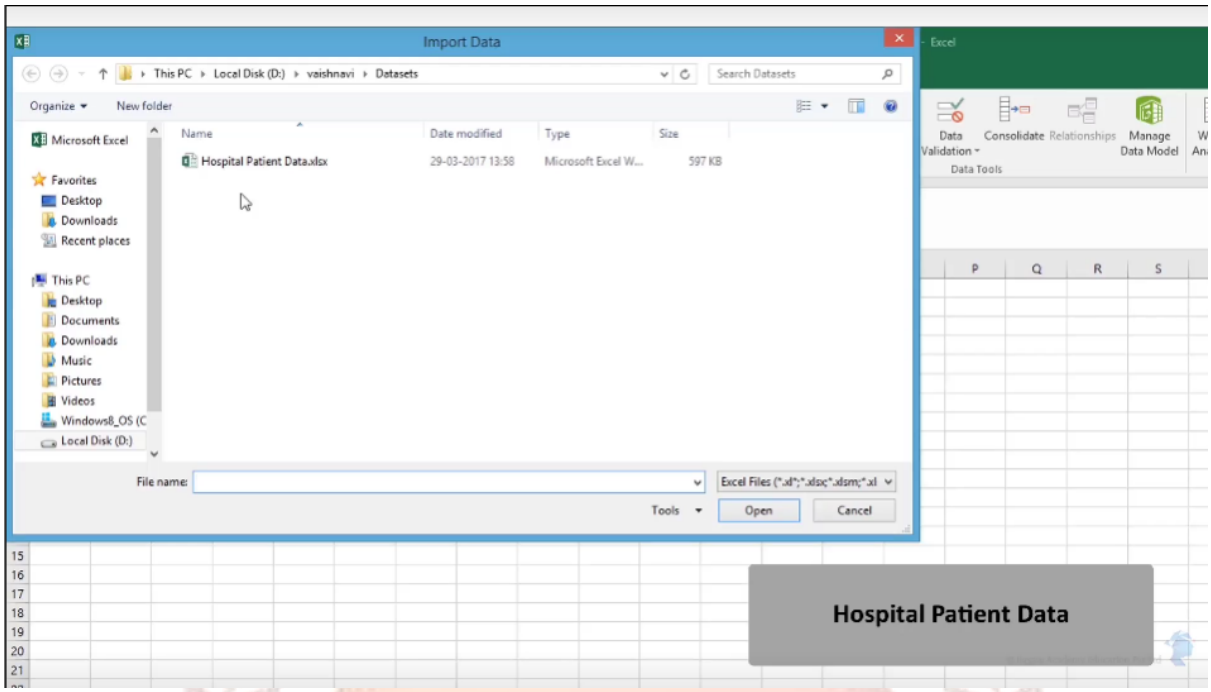
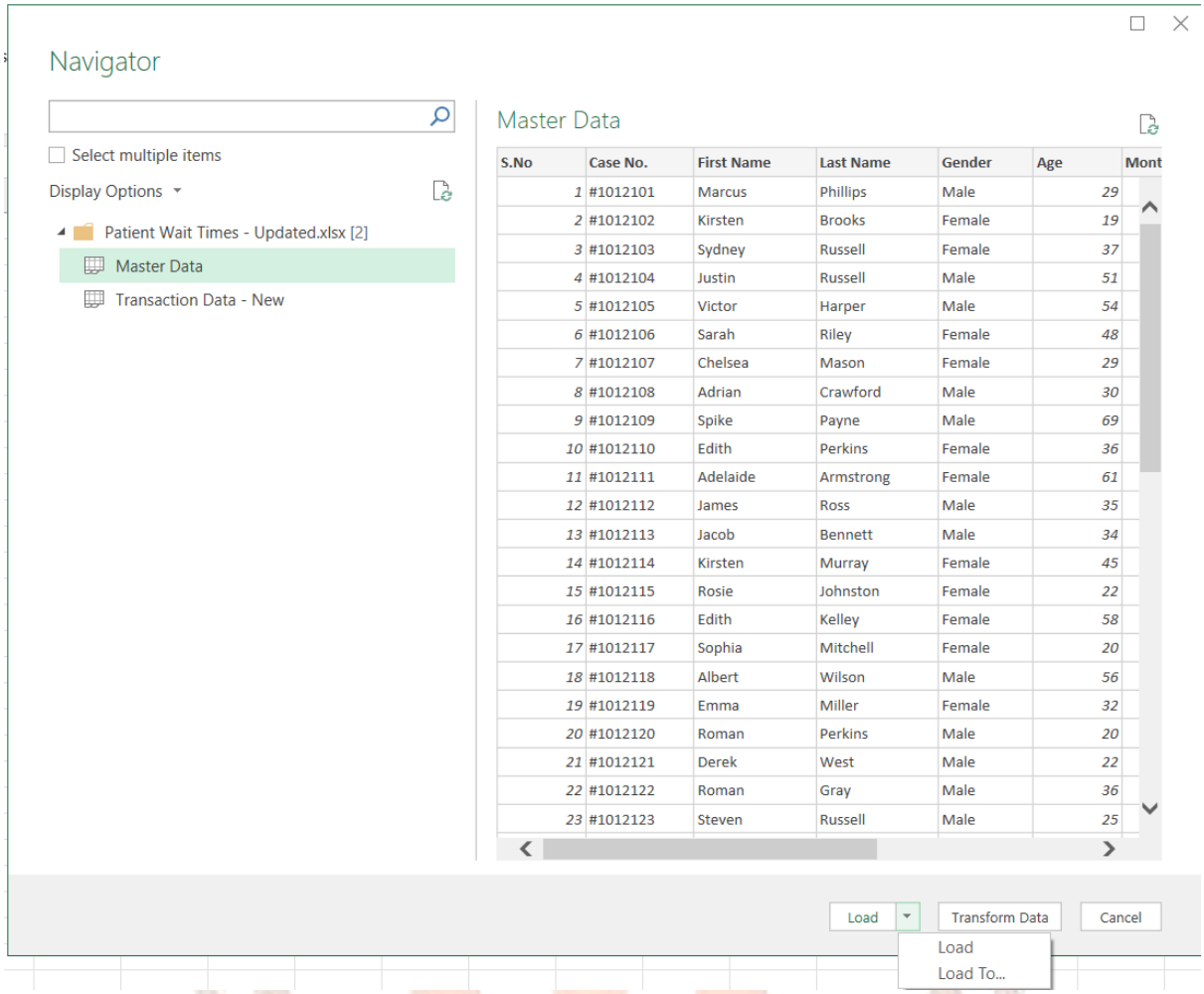


Fig 7: File Data

The selected file data shown in figure 7 is imported to the excel file which opens a navigator window. Select the master data option and click on the load button.



S.No	Case No.	First Name	Last Name	Gender	Age	Mont
1	#1012101	Marcus	Phillips	Male	29	
2	#1012102	Kirsten	Brooks	Female	19	
3	#1012103	Sydney	Russell	Female	37	
4	#1012104	Justin	Russell	Male	51	
5	#1012105	Victor	Harper	Male	54	
6	#1012106	Sarah	Riley	Female	48	
7	#1012107	Chelsea	Mason	Female	29	
8	#1012108	Adrian	Crawford	Male	30	
9	#1012109	Spike	Payne	Male	69	
10	#1012110	Edith	Perkins	Female	36	
11	#1012111	Adelaide	Armstrong	Female	61	
12	#1012112	James	Ross	Male	35	
13	#1012113	Jacob	Bennett	Male	34	
14	#1012114	Kirsten	Murray	Female	45	
15	#1012115	Rosie	Johnston	Female	22	
16	#1012116	Edith	Kelley	Female	58	
17	#1012117	Sophia	Mitchell	Female	20	
18	#1012118	Albert	Wilson	Male	56	
19	#1012119	Emma	Miller	Female	32	
20	#1012120	Roman	Perkins	Male	20	
21	#1012121	Derek	West	Male	22	
22	#1012122	Roman	Gray	Male	36	
23	#1012123	Steven	Russell	Male	25	

Fig 8: Navigator Window

The load option has two sub-options load and load too. The load option allows importing the data into the current worksheet whereas the load to option will allow us to create a new worksheet for the dataset.

Import the dataset into a new workbook by selecting the load to option and selecting **only create connection** option.

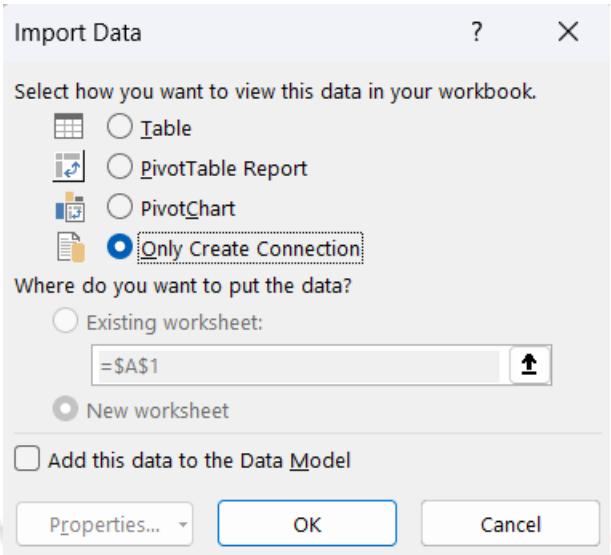


Fig 9: Load to Option

This creates a connection that gets displayed in the workbook area of the excel sheet as shown in figure 10.

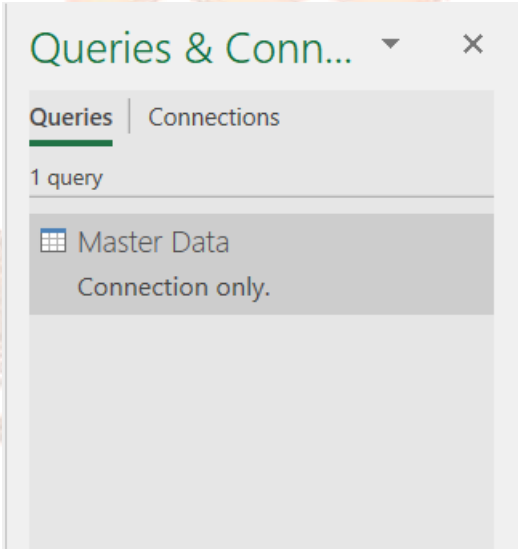
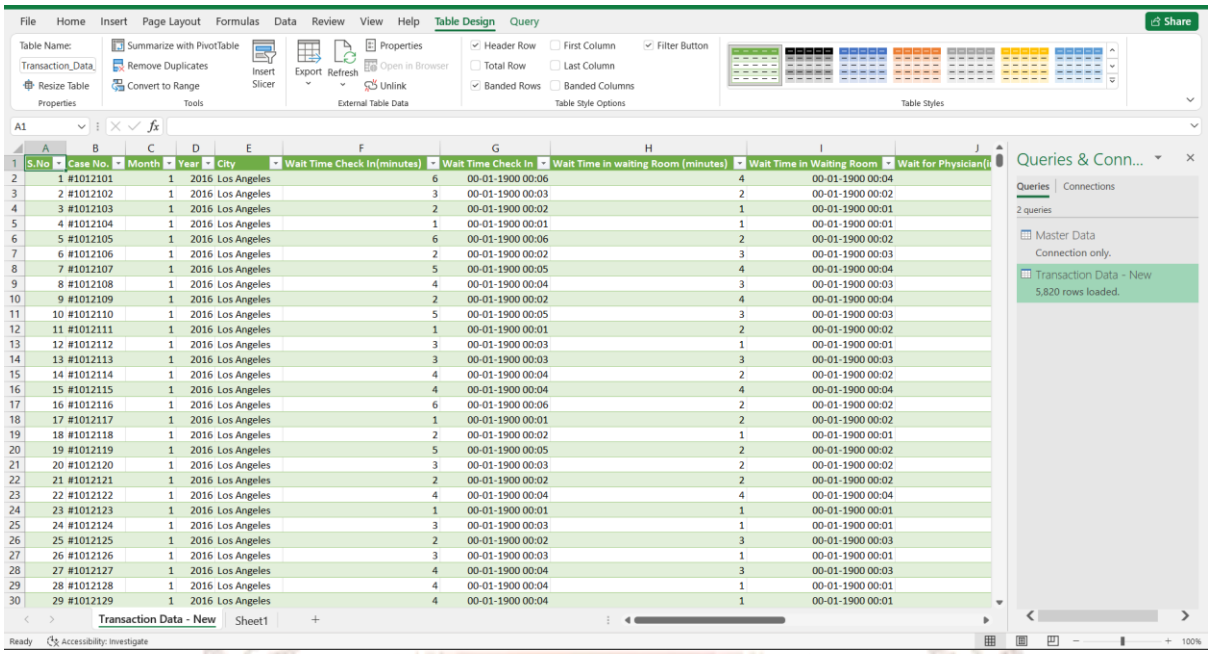


Fig 10: Connection for the Master Data

Now, to import the patient transaction details follow the same steps as per figure 8 and select the patient transaction data and load the data. Below is figure 11 shows the transaction data loaded into the worksheet.



S.No	Case No.	Month	Year	City	Wait Time Check In(minutes)	Wait Time Check In	Wait Time in waiting Room (minutes)	Wait Time in Waiting Room	Wait for Physician(i
1	#1012101	1	2016	Los Angeles	6	00-01-1900 00:06	4	00-01-1900 00:04	
2	#1012102	1	2016	Los Angeles	3	00-01-1900 00:03	2	00-01-1900 00:02	
3	#1012103	1	2016	Los Angeles	2	00-01-1900 00:02	1	00-01-1900 00:01	
4	#1012104	1	2016	Los Angeles	1	00-01-1900 00:01	1	00-01-1900 00:01	
5	#1012105	1	2016	Los Angeles	6	00-01-1900 00:06	2	00-01-1900 00:02	
6	#1012106	1	2016	Los Angeles	2	00-01-1900 00:02	3	00-01-1900 00:03	
7	#1012107	1	2016	Los Angeles	5	00-01-1900 00:05	4	00-01-1900 00:04	
8	#1012108	1	2016	Los Angeles	4	00-01-1900 00:04	3	00-01-1900 00:03	
9	#1012109	1	2016	Los Angeles	2	00-01-1900 00:02	4	00-01-1900 00:04	
10	#1012110	1	2016	Los Angeles	5	00-01-1900 00:05	3	00-01-1900 00:03	
11	#1012111	1	2016	Los Angeles	1	00-01-1900 00:01	2	00-01-1900 00:02	
12	#1012112	1	2016	Los Angeles	3	00-01-1900 00:03	1	00-01-1900 00:01	
13	#1012113	1	2016	Los Angeles	3	00-01-1900 00:03	3	00-01-1900 00:03	
14	#1012114	1	2016	Los Angeles	4	00-01-1900 00:04	2	00-01-1900 00:02	
15	#1012115	1	2016	Los Angeles	4	00-01-1900 00:04	4	00-01-1900 00:04	
16	#1012116	1	2016	Los Angeles	6	00-01-1900 00:06	2	00-01-1900 00:02	
17	#1012117	1	2016	Los Angeles	1	00-01-1900 00:01	2	00-01-1900 00:02	
18	#1012118	1	2016	Los Angeles	2	00-01-1900 00:02	1	00-01-1900 00:01	
19	#1012119	1	2016	Los Angeles	5	00-01-1900 00:05	2	00-01-1900 00:02	
20	#1012120	1	2016	Los Angeles	3	00-01-1900 00:03	2	00-01-1900 00:02	
21	#1012121	1	2016	Los Angeles	2	00-01-1900 00:02	2	00-01-1900 00:02	
22	#1012122	1	2016	Los Angeles	4	00-01-1900 00:04	4	00-01-1900 00:04	
23	#1012123	1	2016	Los Angeles	1	00-01-1900 00:01	1	00-01-1900 00:01	
24	#1012124	1	2016	Los Angeles	3	00-01-1900 00:03	1	00-01-1900 00:01	
25	#1012125	1	2016	Los Angeles	2	00-01-1900 00:02	3	00-01-1900 00:03	
26	#1012126	1	2016	Los Angeles	3	00-01-1900 00:03	1	00-01-1900 00:01	
27	#1012127	1	2016	Los Angeles	4	00-01-1900 00:04	3	00-01-1900 00:03	
28	#1012128	1	2016	Los Angeles	4	00-01-1900 00:04	1	00-01-1900 00:01	
29	#1012129	1	2016	Los Angeles	4	00-01-1900 00:04	1	00-01-1900 00:01	

Fig 11: Patient Transaction Data

The navigator window shown in figure 11 is useful to take a quick look at the available data models that are connected to or being loaded to data model.

In the navigator window, we can also make use of the peek pop-up feature which helps to preview the data by hovering over the data model and we can use the scroll bar to view the data in the model as shown in figure 12.

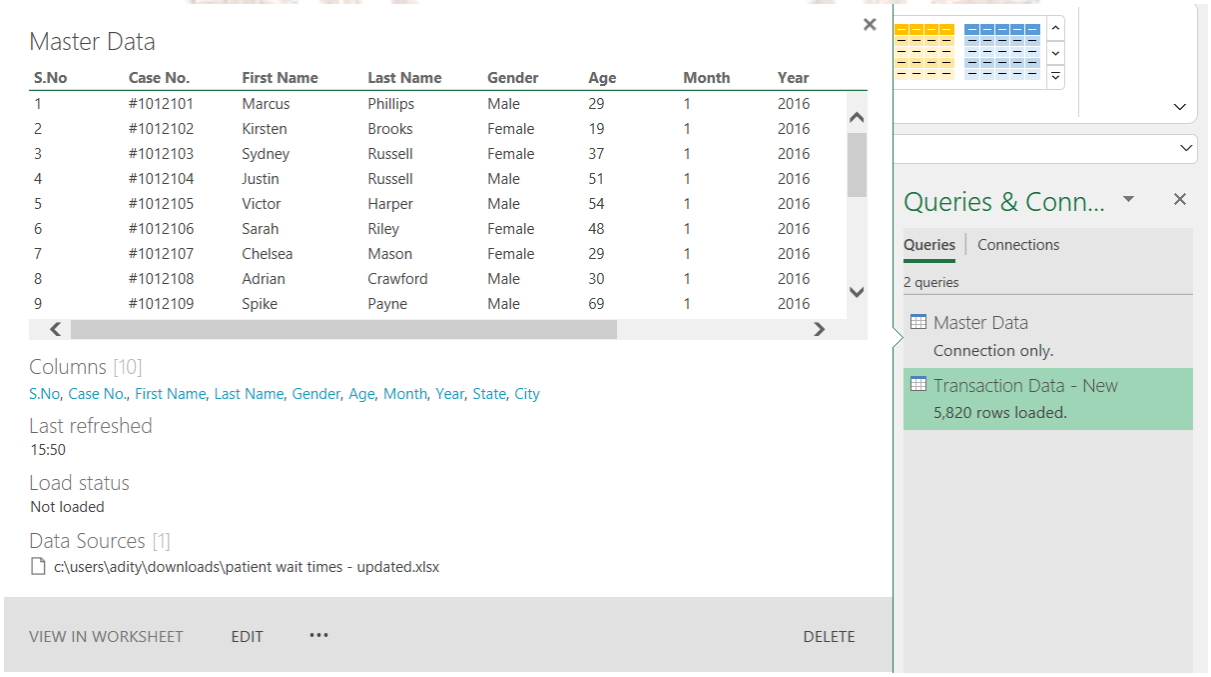


Fig 12: Peek Pop Up Feature

Peek pop-up allows for data discovery without loading unnecessary data. Note that each load is stored as an individual query and it appears on the navigator window as it is being connected to other data sources.

In the navigator window, we can right-click on the data model we can access many options which allow us to copy, edit, load, or perform many such operations as shown in figure 13.

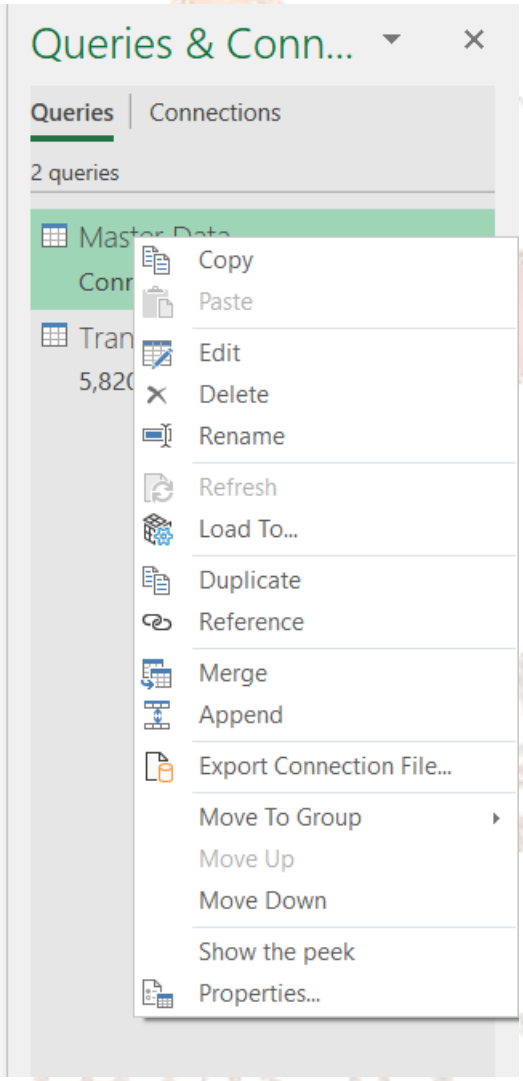


Fig 13: Options Available to Make Changes to the Data

Connecting to Data from Database

To load data from a database:

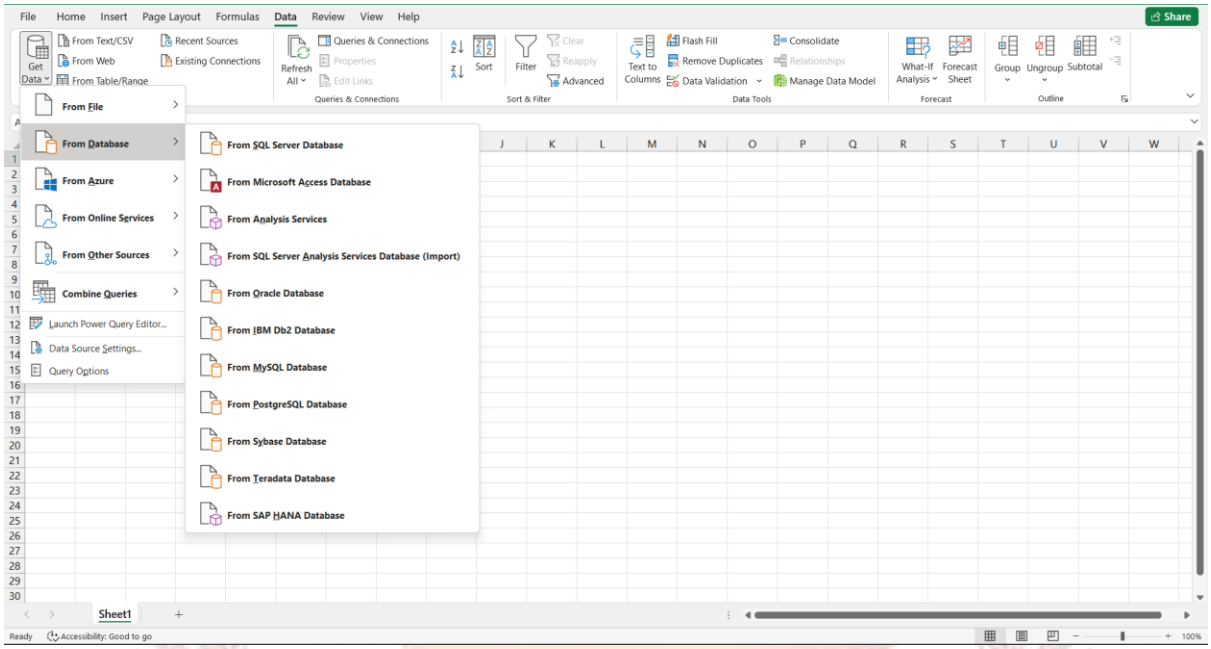


Fig 14: Data Collection from Database

Select the required database option as shown in figure 14.

For example, consider the SQL server database. Select the SQL database option which will provide a window where we need to enter the server name and details for any specific query or relationship column required as shown in figure 15.

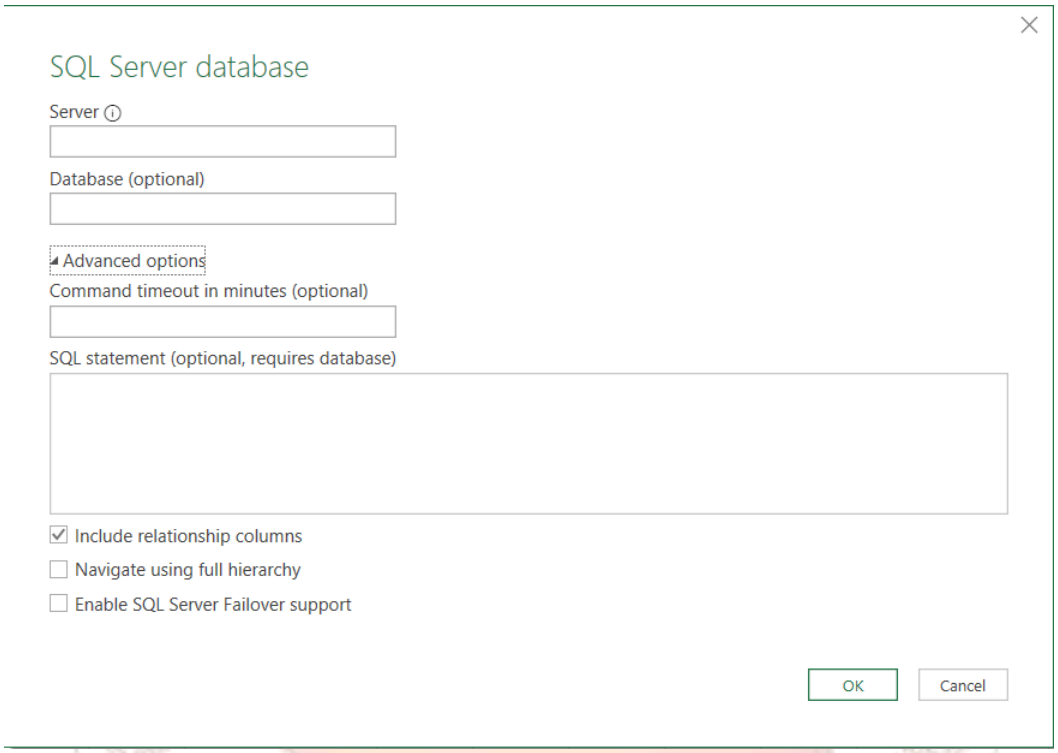


Fig 15: SQL Server Database Connection

Connecting to XML File

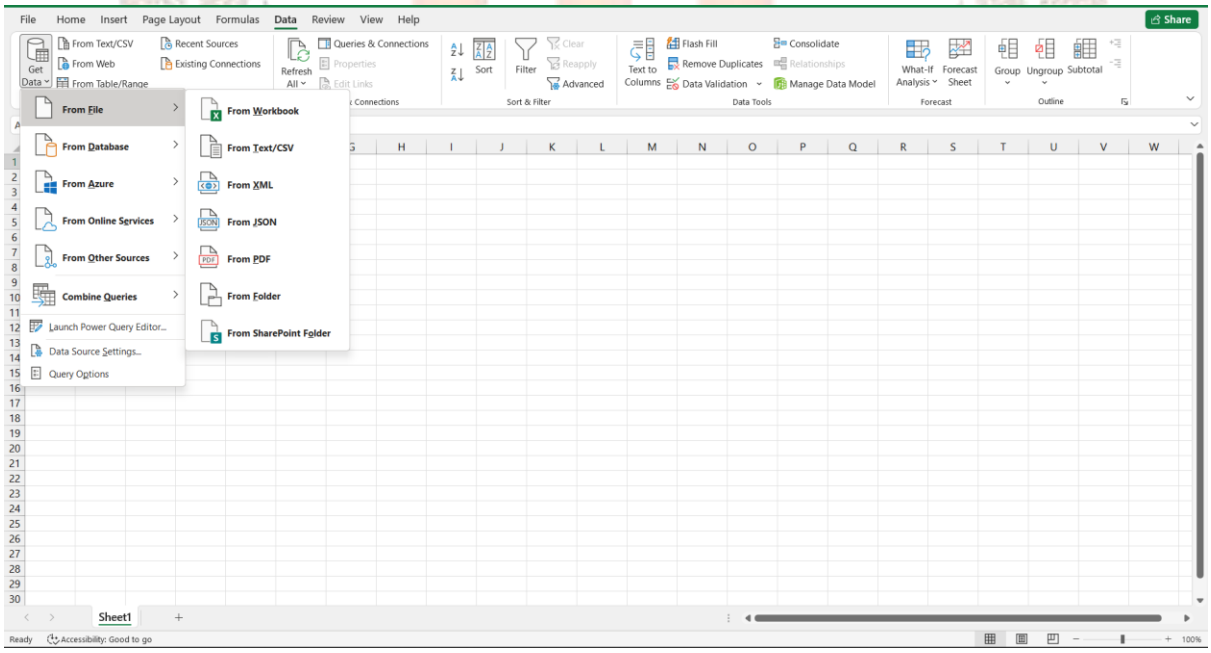


Fig 16: Data from File Option

From the collection of data from file, option choose the XML option. Select the patients.xml file which contains the patient's data in an XML format and import the data as shown below in figure 17.

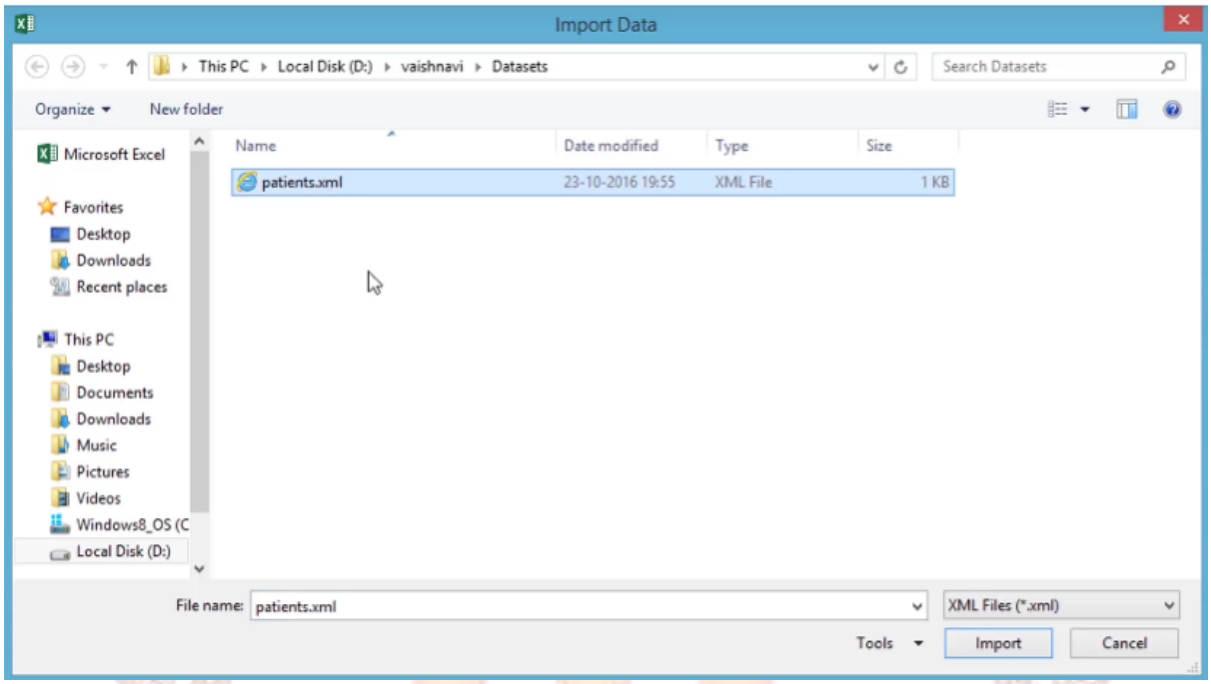


Fig 17: Patients.xml File

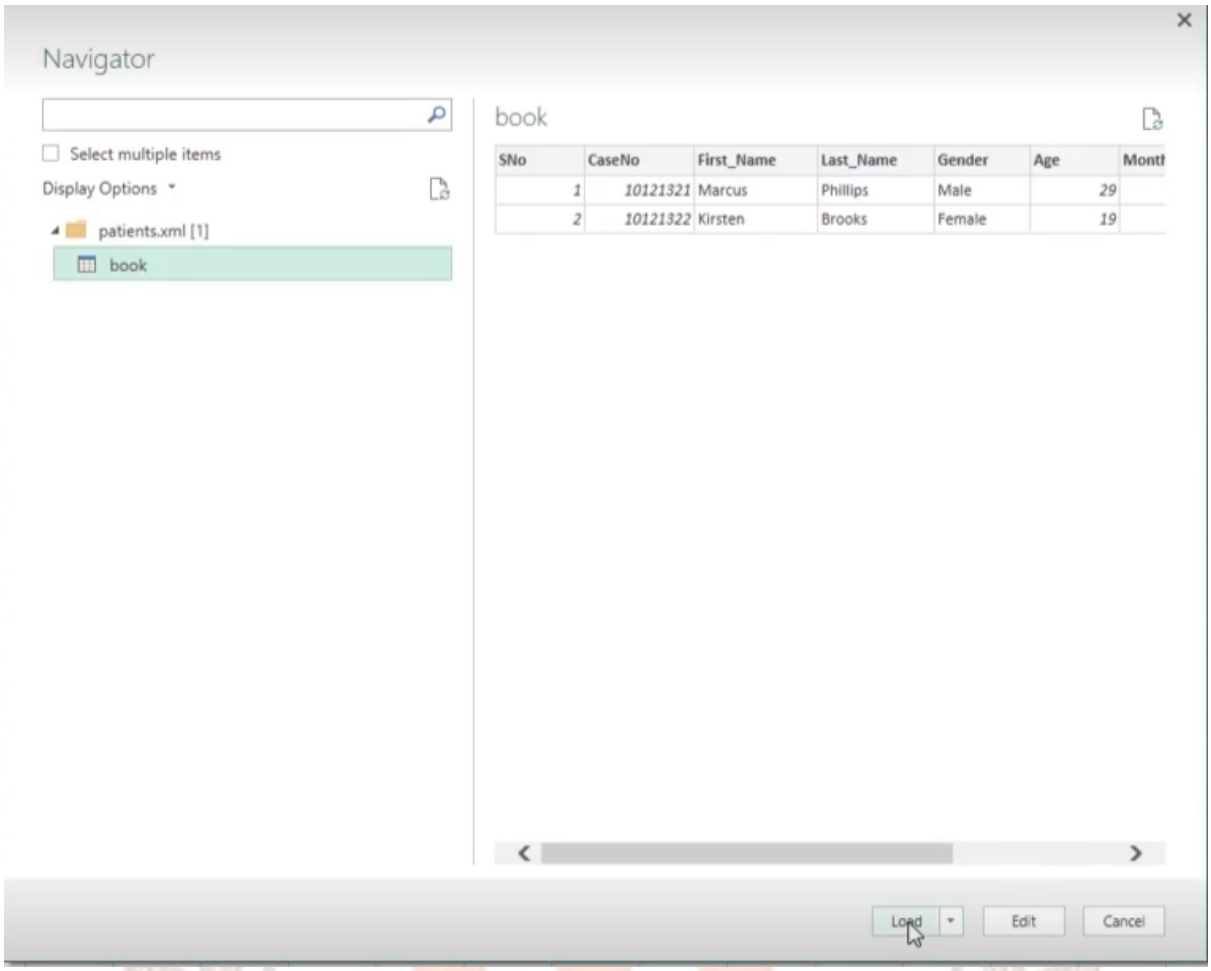


Fig 18: Patients Data from the XML File

Select the book option and load the data as shown in figure 18. This will load the data from the XML file into the excel data model as shown in figure 19.

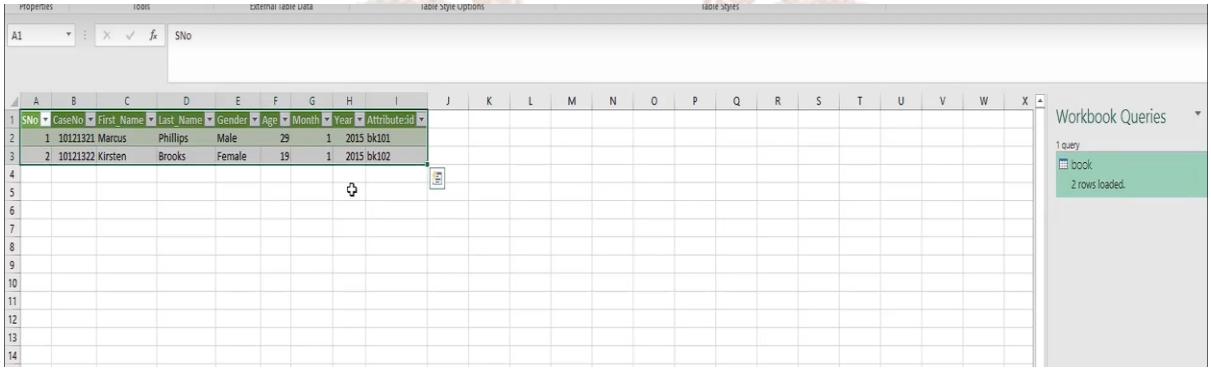


Fig 19: Patients Data Loaded into the Excel Data model

Figure 19 also shows the query connection which has been established to the XML schema on the right side of the excel sheet in the workbook queries window.

Connecting to Web and Social Media Files:

Advance data discovery can be done by looking at data sources outside or from available data sources by connecting with external data . Consider connecting to an external web source for a list of hospitals in a specific region (patient data is available).

To connect to web source to obtain data select the new query option in Data tab and under 'from other sources' select the from web option as shown in figure 20.

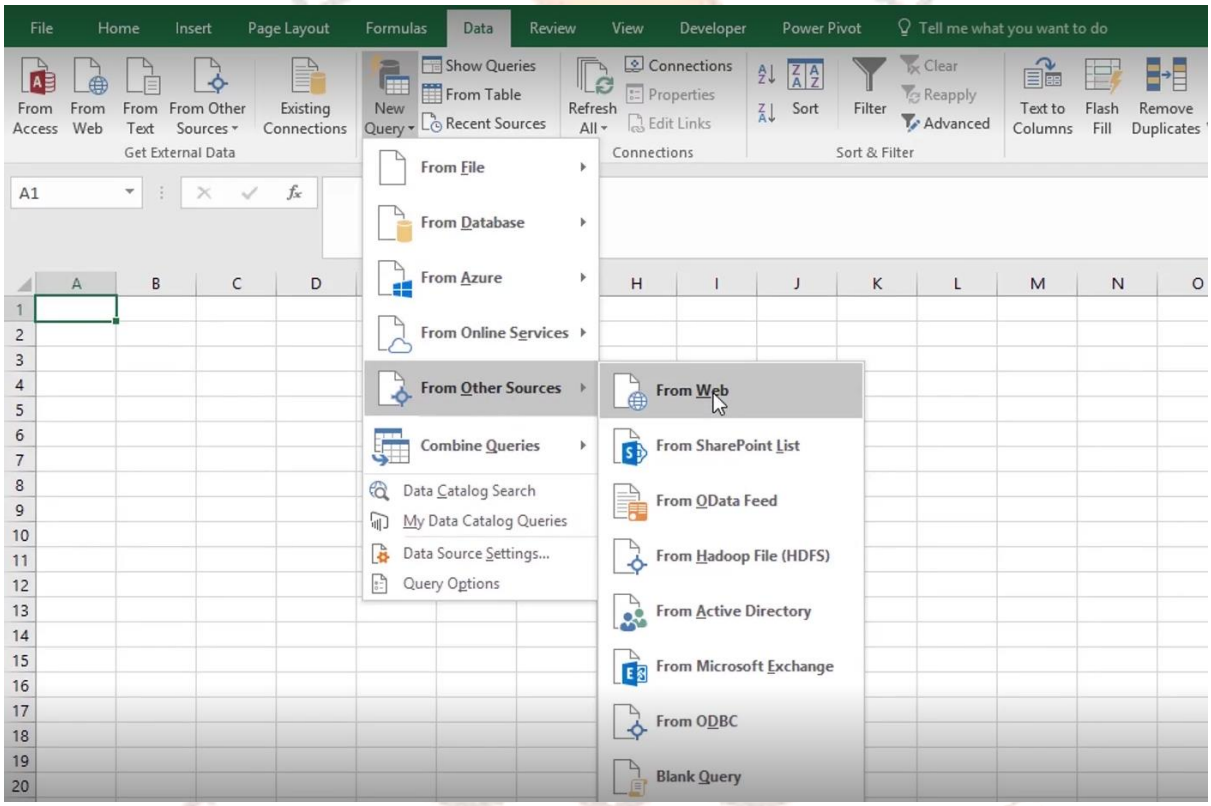


Fig 20: Connecting to WEB Source

Following the steps shown for figure 20 a pop-up window will appear on the worksheet where we need to enter the URL in the empty field as shown in figure 21.

Fig 21: Connecting to Web Using URL

For this scenario we are loading the hospital data from the California region as shown in figure 21. Click ok to go to the navigator window where the HTML table element is loaded as shown in figure 22.

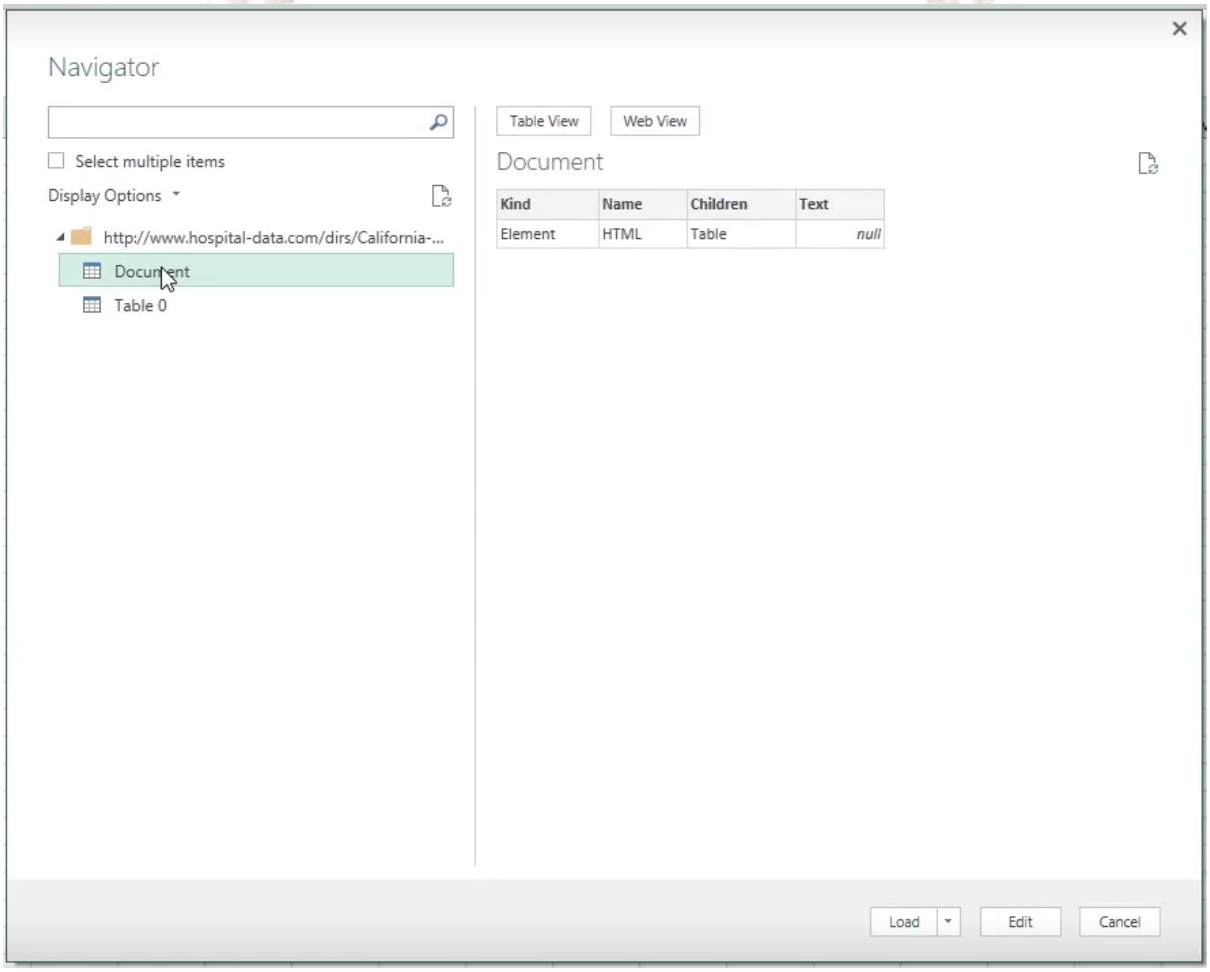


Fig 22: HTML Element

Figure 23 shows a data table which consists of 3 columns (maps, name, city), since we do not require map column, we can remove the column using the query editor

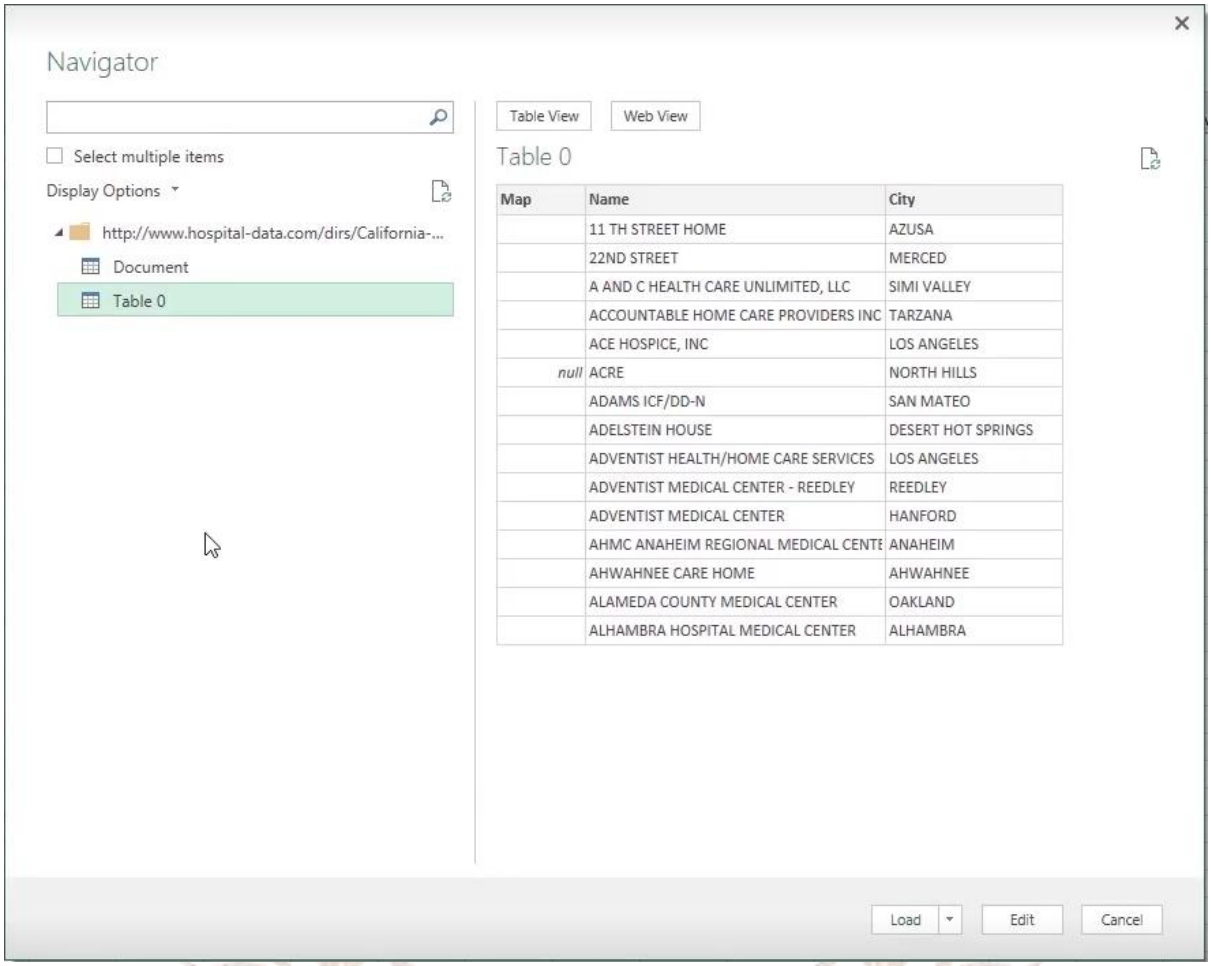


Fig 23: Data Table

Since we need to remove the map column click on edit which will open the query editor window where we can remove any unnecessary columns.

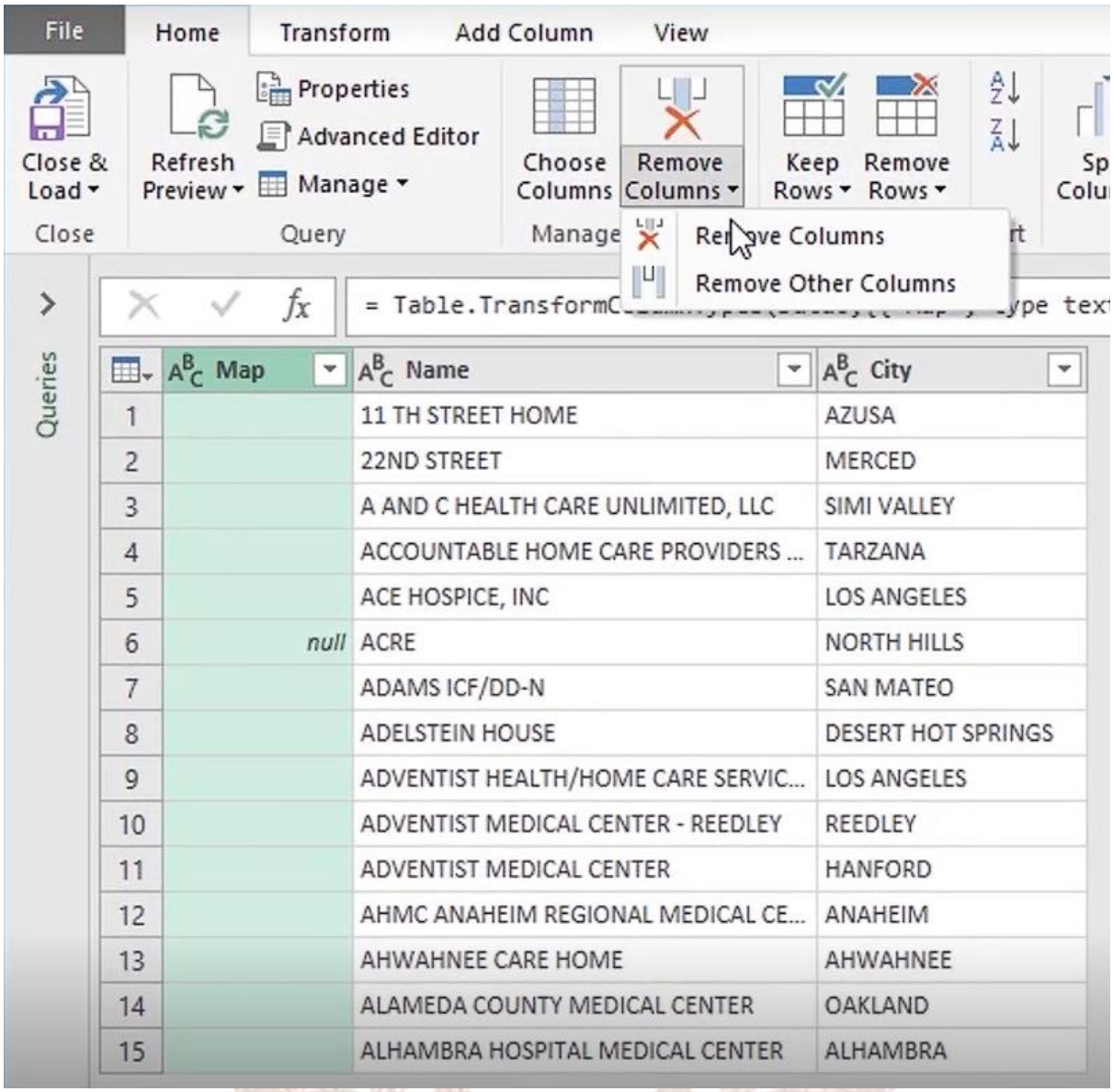


Fig 24: Query Editor

Select the required column and click on remove columns option from the top menu as shown in figure 24. This will delete the maps column and only the name and city column will remain which is required for the data model.

Alternate Method to Connect to Web Data:

Alternate method to connect to web data is to use formula in the query editor to connect to a web source. To open the query editor, select the 'blank query' option under from other sources in new query ribbon from data tab as shown in figure 25.

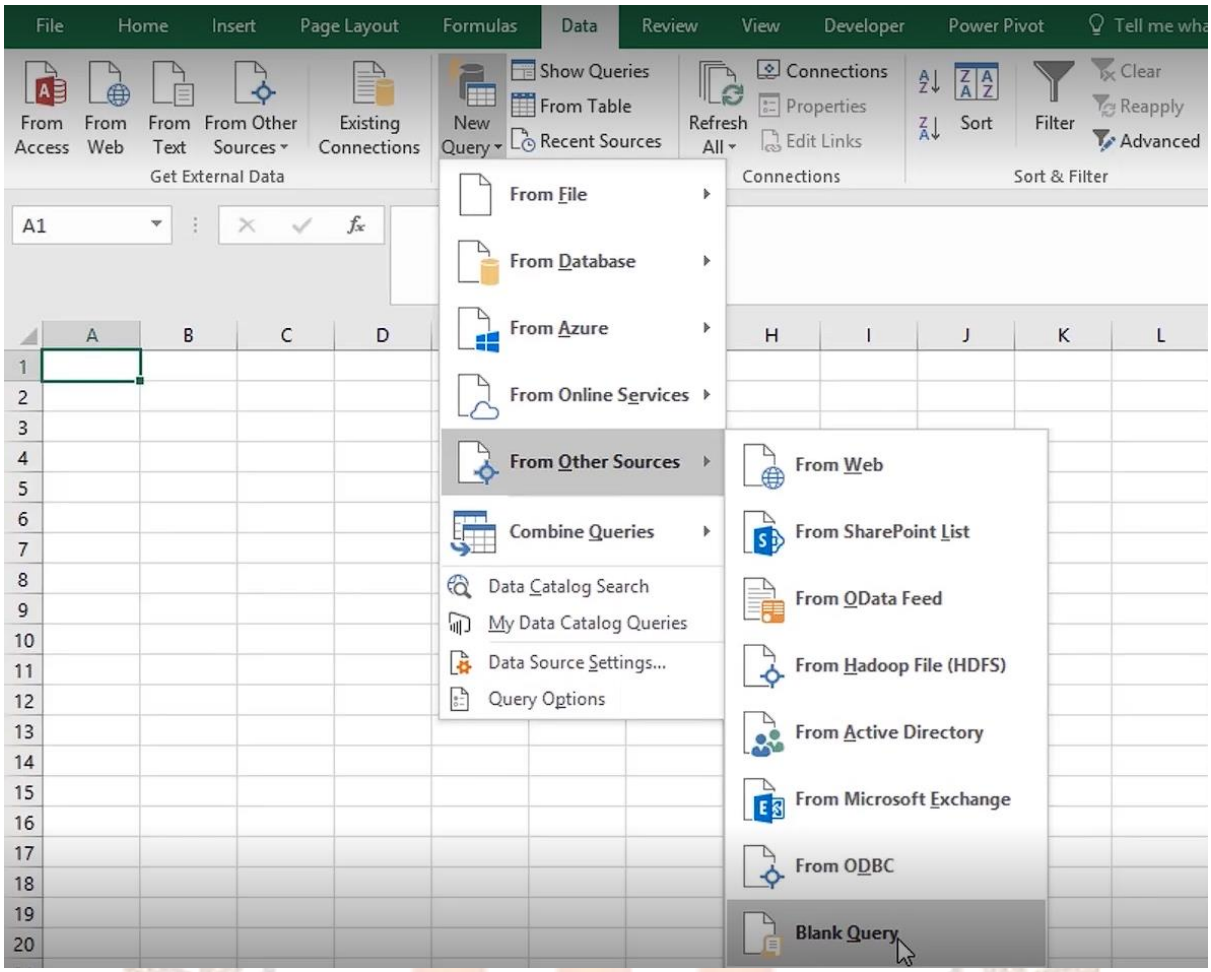


Fig 25: To Open Editor Window

This will open a blank query window where we need to enter the formula in the empty field provided. Enter the formula as shown in figure 26.

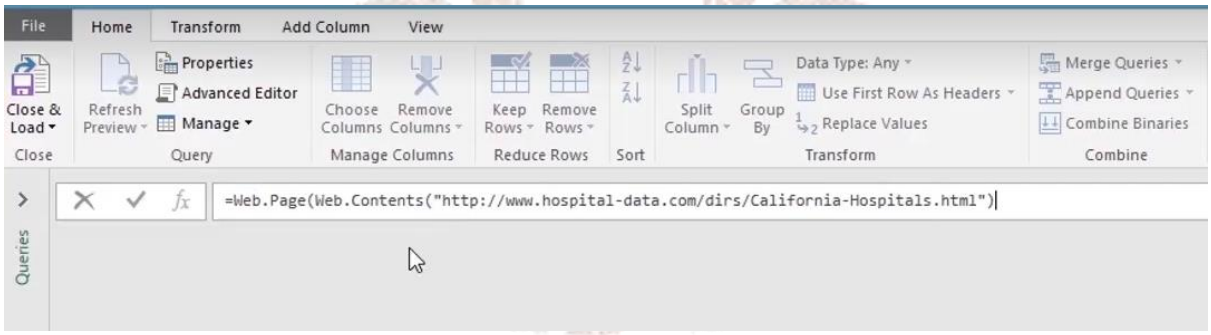


Fig 26: Formula to Connect to Web Data

Note that the above shown formula is case sensitive.

Once the formula is entered the HTML table structure will be loaded and on expanding the data column, we will be able to see the various data attributes of the data table as shown in figure 27.

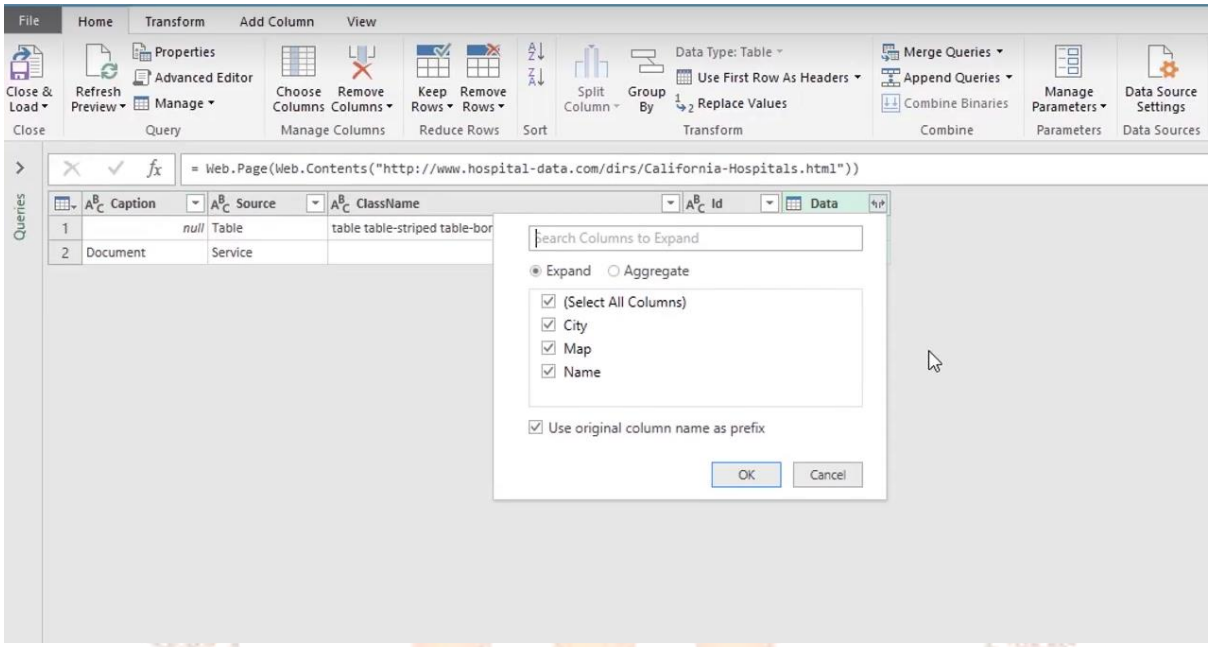
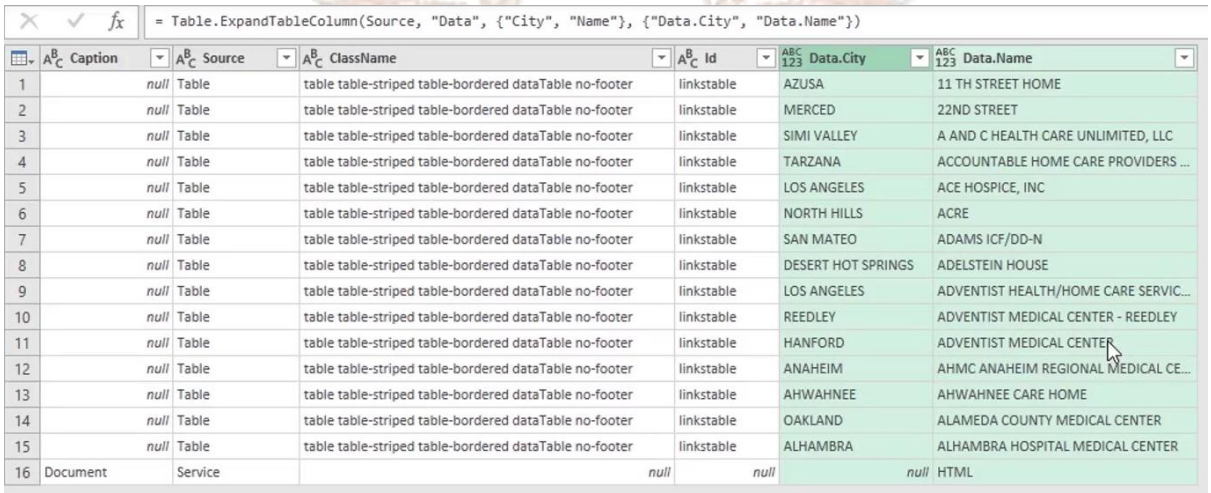


Fig 27: Data Loaded from the Formula

In figure 27 we can see the attributes available in the data table, here we can choose which data we need to display and as we do not require the maps data uncheck the maps attribute and select ok.

This will show the data along with the city and name as displayed in figure 28.



= Table.ExpandTableColumn(Source, "Data", {"City", "Name"}, {"Data.City", "Data.Name"})

	Caption	Source	ClassName	Id	Data.City	Data.Name
1	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	AZUSA	11 TH STREET HOME
2	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	MERCED	22ND STREET
3	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	SIMI VALLEY	A AND C HEALTH CARE UNLIMITED, LLC
4	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	TARZANA	ACCOUNTABLE HOME CARE PROVIDERS ...
5	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	LOS ANGELES	ACE HOSPICE, INC
6	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	NORTH HILLS	ACRE
7	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	SAN MATEO	ADAMS ICF/DD-N
8	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	DESERT HOT SPRINGS	ADELSTEIN HOUSE
9	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	LOS ANGELES	ADVENTIST HEALTH/HOME CARE SERVIC...
10	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	REEDLEY	ADVENTIST MEDICAL CENTER - REEDLEY
11	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	HANFORD	ADVENTIST MEDICAL CENTER
12	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	ANAHEIM	AHMC ANAHEIM REGIONAL MEDICAL CE...
13	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	AHWAHNEE	AHWAHNEE CARE HOME
14	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	OAKLAND	ALAMEDA COUNTY MEDICAL CENTER
15	null	Table	table table-striped table-bordered dataTable no-footer	linkstable	ALHAMBRA	ALHAMBRA HOSPITAL MEDICAL CENTER
16	Document	Service	null	null	null	HTML

Fig 28: Displayed Data for Selected Attributes

Connecting to Online Data Sources:

Consider a scenario where we need to analyse the social media sentiment of patients, for this purpose we are required to connect to the patients Facebook page for this scenario. For the purpose of this demo, we have created a Facebook page as seen in figure 29.

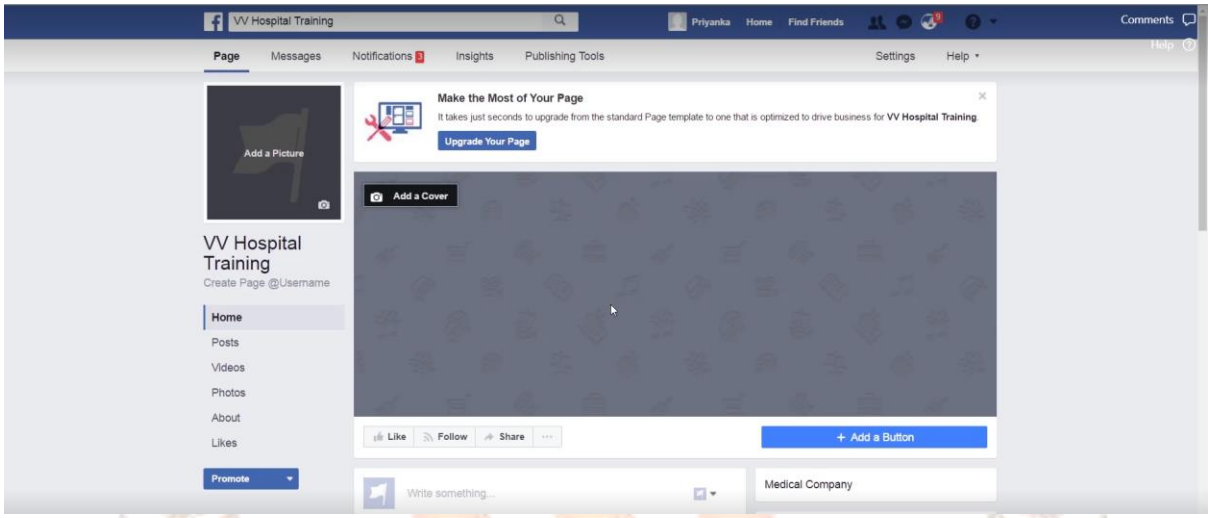


Fig 29: Demo Facebook Page

In figure 29 we have created a demo page VV Hospital Training for the purpose of the scenario. Here we can extract the data like posts comments likes etc. Now, in order to connect to the above Facebook data, click on new query and under online services select 'from Facebook' as shown in figure 30.

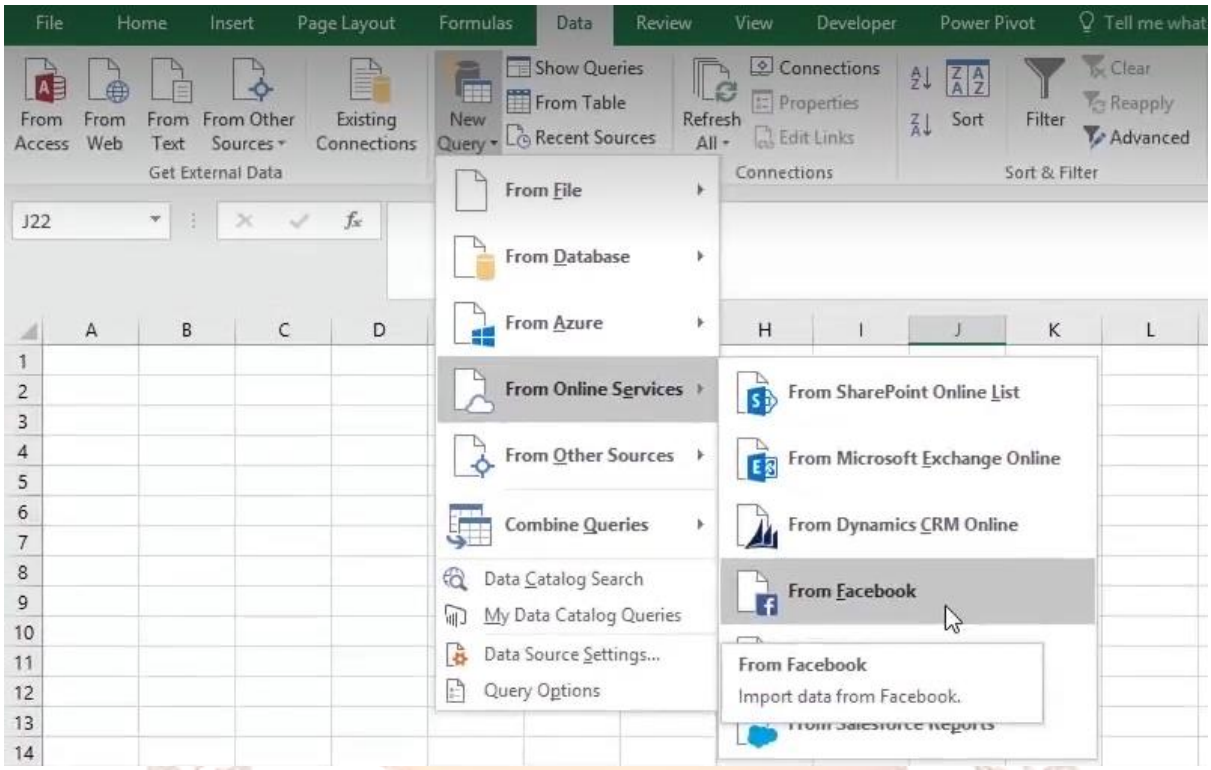


Fig 30: Connecting to Facebook Data

Following the above step will open a Facebook window which requests an object or a connection in the Facebook graph. Facebook graph is the primary way for any external app to talk with Facebook. Using the Facebook graph, we can access all the information we require that can be handled by excel. To access the required object id open the Facebook page in the browser and copy the numbers which appears after the name in the URL at the end of page as shown in figure 31.

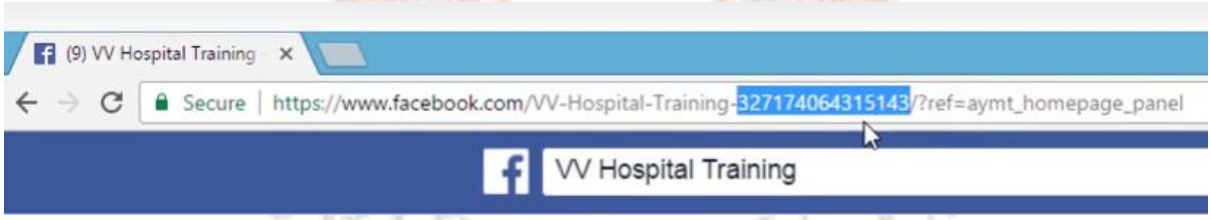


Fig 31: URL of Demo Page

The highlighted value in figure 31 is the object id of the demo page which we need in order to establish the connection from excel.

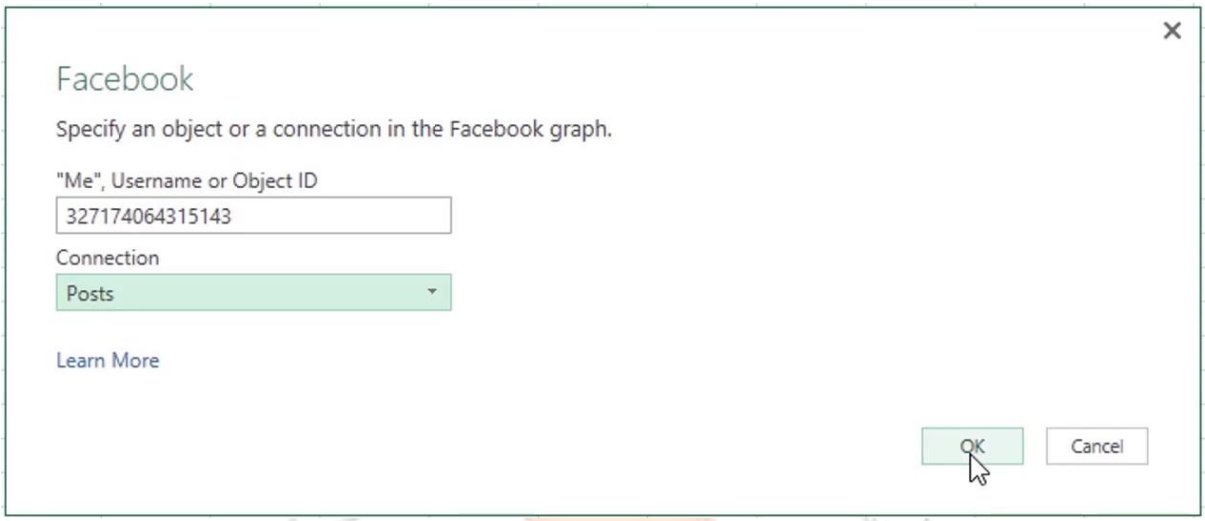


Fig 32: Facebook Connection Window

The above figure shows the pop-up window which appears following the steps in figure 30. The object id is entered which we obtained from the URL and the connection attributes we need here is posts. Clicking on ok will retrieve all information related to the page as shown in figure 33.

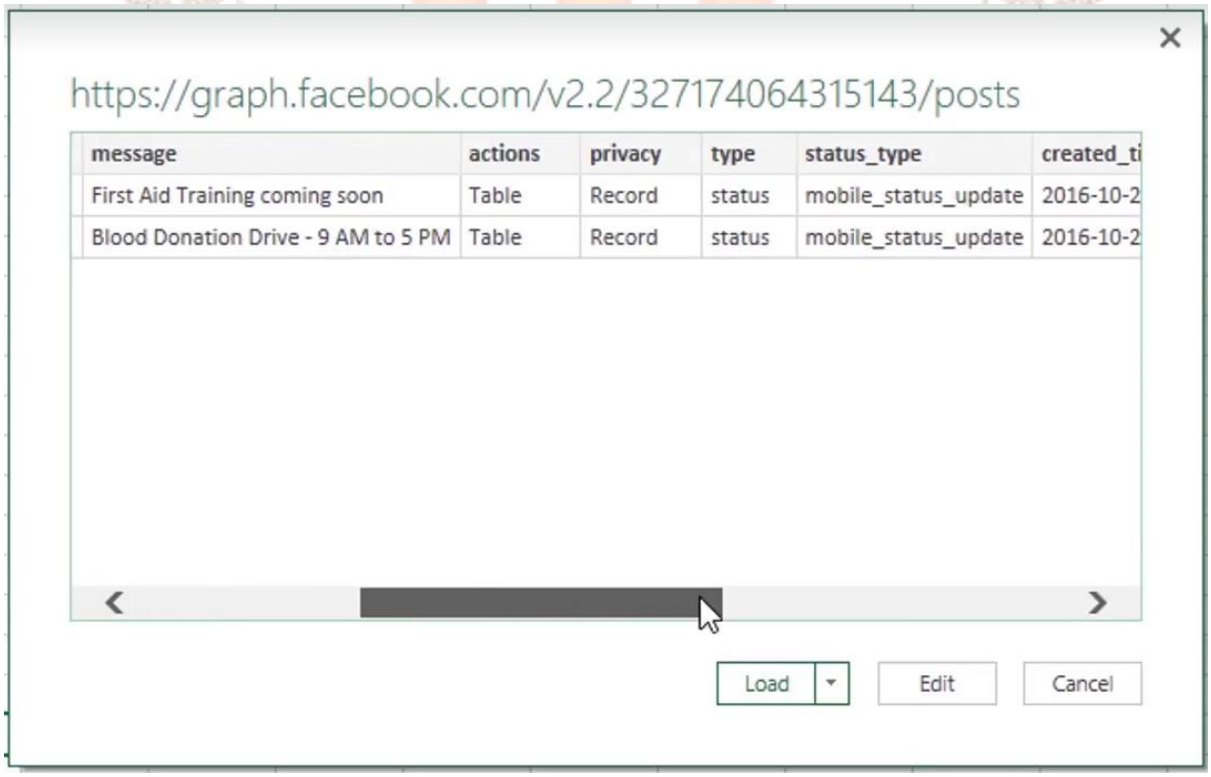


Fig 33: Retrieved Information from the Demo Page

Click on edit will open the query editor and we can use the header to see additional information that is available for that attribute as shown in figure 34.

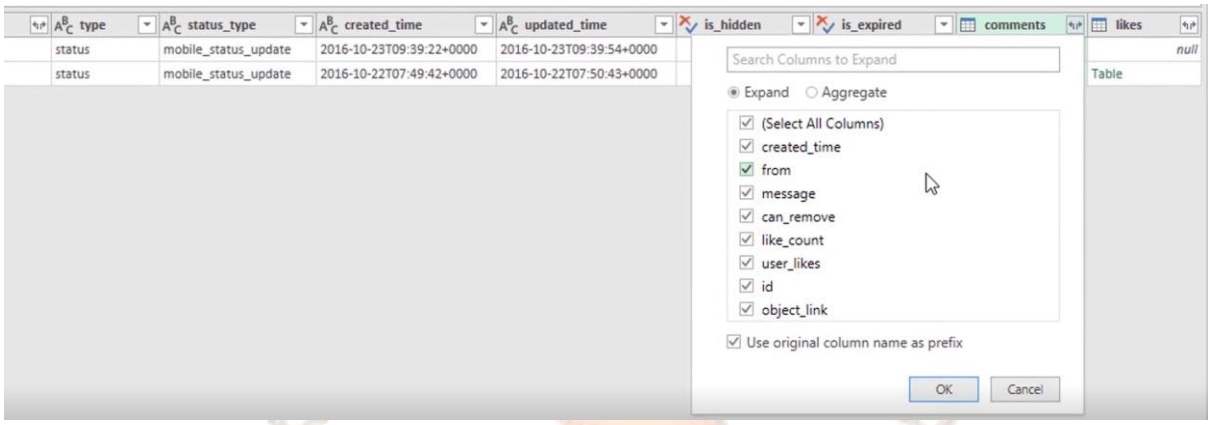


Fig 34: Comments Data Attributes

Figure 34 shows the data attributes that can be extracted from the comments data. By clicking on ok the data from the comments will be added to the table as shown in figure 35.

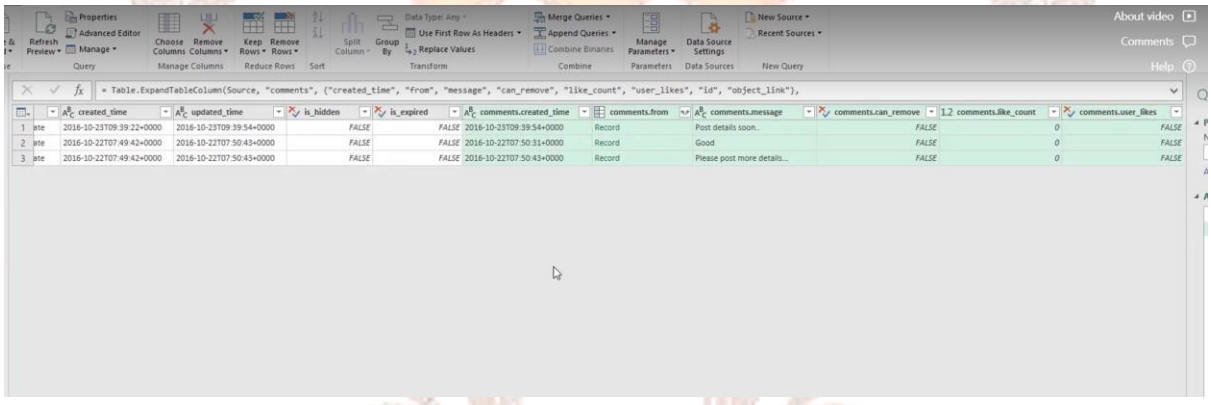


Fig 35: Extracted Data from the Comments Column

The above shown steps shows how to connect to an online Facebook source to extract data, the same data can also be extracted using a similar formula by clicking on a new blank query option (refer figure 25). For this we need to choose new query from data tab, then clicking on from other sources and then on blank query, will open power query editor window.

The below figure 36 shows the formula used to connect to the Facebook id.

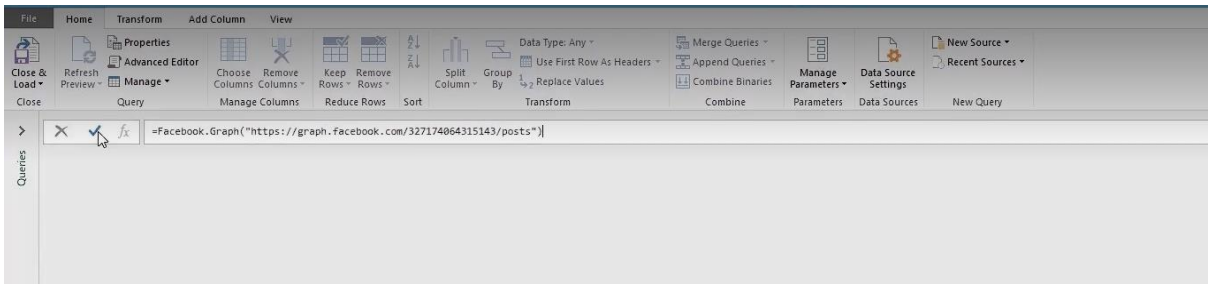


Fig 36: Formula to Connect to Facebook

Once we get the power query editor window, formula is required to enter to extract the post from Facebook graph id . By clicking enter a similar window will pop-up with all the data that appear on that web page, here the comments data can also be extracted using the same steps from figure 34 and the resulting data will be the same as shown in figure 35. Here we have discussed examples to connect to various data sources.

Summary

In this topic we discussed,

- How to load data into the Excel model
- Various methods to connect and load data into the Excel sheet.
- Various methods to connect to Web data source.